

Heat Stroke Risk Prediction - Project Report

Overview

This report summarizes an end-to-end binary classification project for predicting heat stroke risk using physiological, environmental, and lifestyle-related factors.

Dataset

The dataset contains 609 patient records with 27 attributes including body temperature, rectal temperature, environmental temperature, heat index, humidity, age, physical activity, hydration, BMI, blood pressure, and other clinical variables. The target variable is Heat stroke (0 = No Heat Stroke, 1 = Heat Stroke).

Preprocessing

Target values were cleaned and converted to numeric format. Median imputation was used where necessary. Numerical features were standardized using StandardScaler before model training.

Models

Logistic Regression and Random Forest Classifier were trained and evaluated.

Evaluation

Performance was measured using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and Classification Report.

Observations

Temperature-related variables and Heat Index showed strong association with heat stroke occurrence. Correlation analysis highlighted relationships among physiological variables.

Conclusion

The project demonstrates a complete machine learning workflow including preprocessing, exploratory data analysis, model training, evaluation, and prediction. Future improvements include hyperparameter tuning and additional ensemble models.

Feature Visualizations



